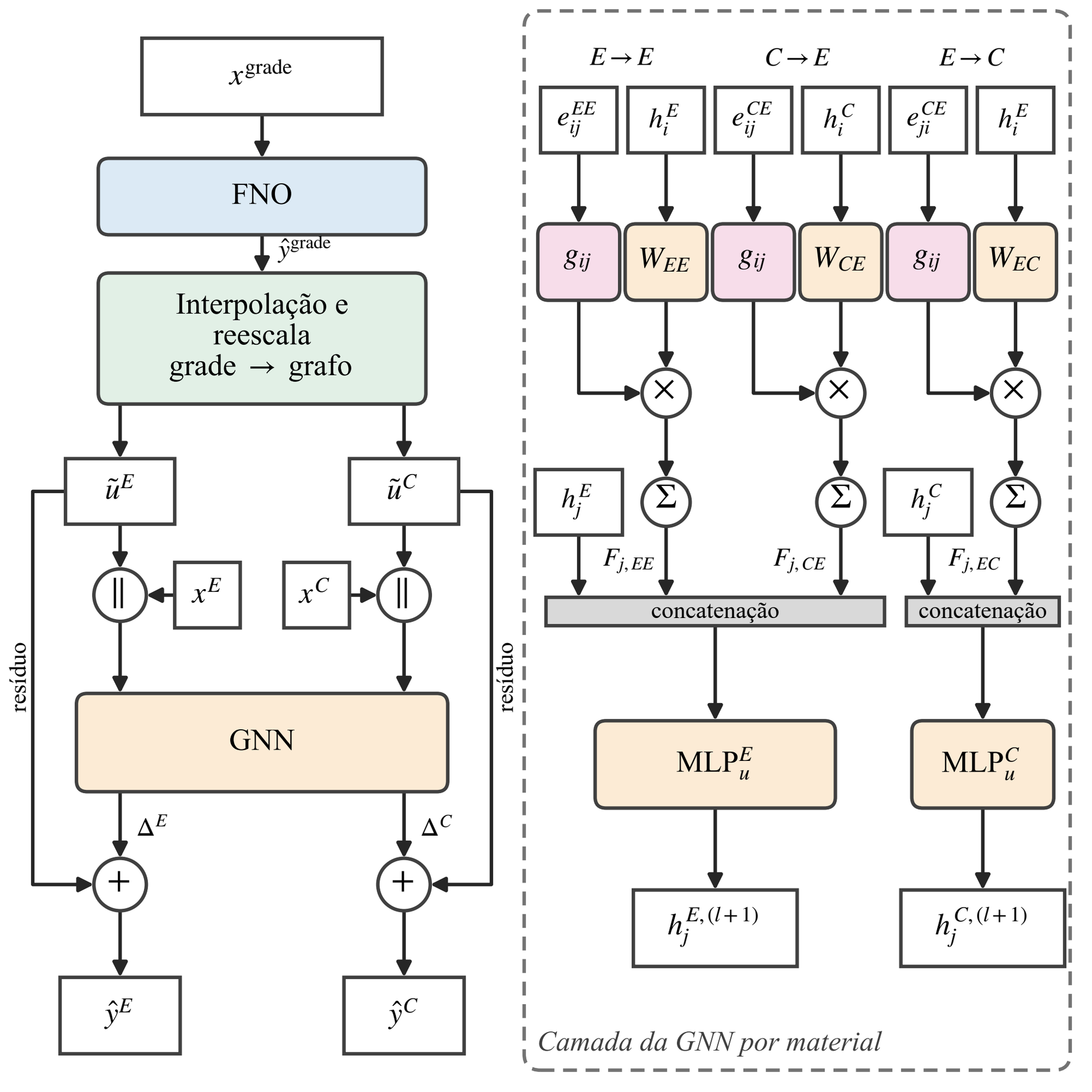
x^{grade}
FNO
$\hat{y}^{\text{grade}}$
Interpolação e reescala grade → grafo
$\tilde{u}^E$
$\tilde{u}^C$
x^E
x^C
resíduo
resíduo
GNN
Δ^E
Δ^C
$\hat{y}^E$
$\hat{y}^C$
$E \to E$
$C \to E$
$E \to C$
e_{ij}^{EE}
h_i^E
e_{ij}^{CE}
h_i^C
e_{ji}^{CE}
h_i^E
g_{ij}
W_{EE}
g_{ij}
W_{CE}
g_{ij}
W_{EC}
h_j^E
h_j^C
$F_{j,EE}$
$F_{j,CE}$
$F_{j,EC}$
concatenação
concatenação
MLP_u^E
MLP_u^C
$h_j^{E,(l+1)}$
$h_j^{C,(l+1)}$
Camada da GNN por material